

Question ID: 4d3e3c52

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Central Ideas and Details	Hard

Question

In a paper about p-i-n planar perovskite solar cells (one of several perovskite cell architectures designed to collect and store solar power), Lyndsey McMillon-Brown et al. describe a method for fabricating the cell’s electronic transport layer (ETL) using a spray coating. Conventional ETL fabrication is accomplished using a solution of nanoparticles. The process can result in a loss of up to 80% of the solution, increasing the cost of manufacturing at scale—an issue that may be obviated by spray coating fabrication, which the researchers describe as “highly reproducible, concise, and practical.”

What does the text most strongly suggest about conventional ETL fabrication?

Answer

- A. It is less suitable for manufacturing large volumes of planar p-i-n perovskite solar cells than an alternative fabrication method may be.
- B. It is more expensive when manufacturing at scale than are processes for fabricating ETLs used in other perovskite solar cell architectures.
- C. It typically entails a greater loss of nanoparticle solution than do other established approaches for ETL fabrication.
- D. It is somewhat imprecise and therefore limits the potential effectiveness of p-i-n planar perovskite solar cells at capturing and storing solar power.

Correct Answer: A

Rationale

Choice A is the best answer. Conventional solar cell fabrication increases “the cost of manufacturing at scale,” but spray coating might get rid of that problem.

Choice B is incorrect. This is not completely supported by the text. While it’s true that conventional ETL fabrication is expensive at scale, there’s nothing in the text that mentions other perovskite solar cell architectures. Choice C is incorrect. This choice does not match the text. Only one conventional method of ETL fabrication is described, so we can’t compare the solution loss in this method to that of other conventional methods. Choice D is incorrect. This choice isn’t supported by the text. The text never suggests that the effectiveness of solar cells changes based on their method of fabrication.